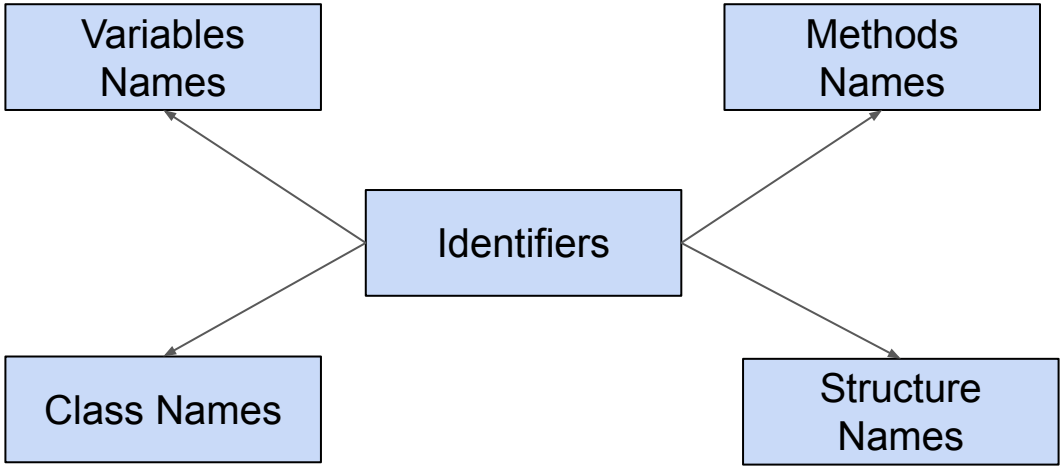
Introduction to Variables and Datatypes

Identifiers

- Name assigned to a particular entity in a program.
- Can be a variable name, method name, class name, structure name etc.

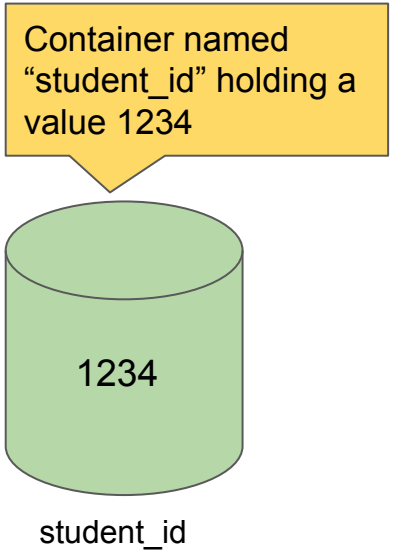


Variables

- A variable is a container which is used to store data.
- A kind of identifier.
- Each variable has a specific data type.

```
int   student_id   =   1234;
```

Datatype **Variable_Name** **Value**



Variables Initialization

- **Syntax**:

  <datatype> <variable_name> = <value> <format_specifiers>

  [Format specifiers only for double and float data type]

- **Example**:

```
int  number = 25;
char var = ‘H’;
double value = 2.56d;
```

Variable Naming Conventions

- Must begin with letters (A-Z, a-z), dollar sign ($) or underscore (_).

  Ex: int number = 90;

- Can not start with numbers or special characters.

  Ex: int %name = 88;

  Ex: int 1name = 70;

  Wrong ✗

- Can have numbers except at the start (0-9).

  Ex: int num1 = 10;

Variable Naming Conventions (Contd)

- Case sensitive

  Num1 and num1 not same

- Can not contain white spaces

- Can use **camel casing** (studentName) or **snake casing** (student_id) for multiple words in a variable

  Ex: int given number = 20; Wrong

  Ex: int given_number = 20; Correct

  Ex: int givenNumber = 20; Correct

Variable Naming Conventions (Contd)

- Should be meaningful.
- Single character variable names should be avoided.
- Variable names must not be a Java reserved keyword.

abstract	assert	boolean	synchronized	byte	case	catch	char	class
continue	default	do	double	else	enum	extends	final	finally
float	for	if	implements	import	int	long	native	new
null	package	private	instanceOf	public	return	short	static	strictfp
super	switch	break	interface	this	throw	throws	transient	try
void	volatile	while	protected	const	goto	true	false	null

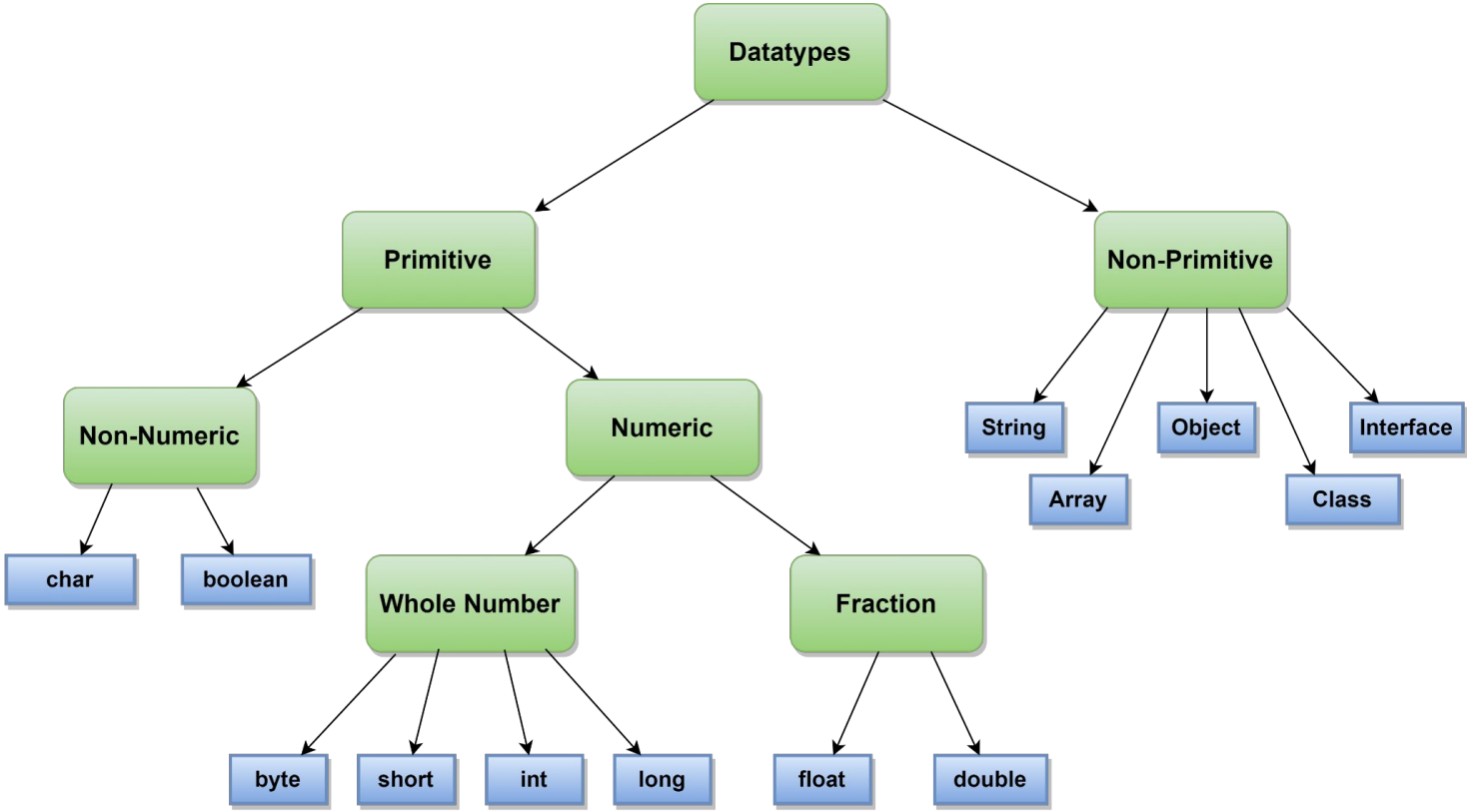
Fig: Reserved keywords in Java

Data Types

- Each variables in Java must of a specific data type.
- There are major two types of data in Java:

    1. **Primitive Data Types:** Holds a value, specifies the size and type of a variable.
    2. **Non-Primitive Data Types:** Holds a reference to an object or location.

Data Types (Contd)



Data Types (Contd)

Data Type	Significance	Range
byte	8-bit integer	-128 to 127
short	Short integer	-32,768 to 32,767
int	Integer value	–2,147,483,648 to 2,147,483,647
long	Long integer	–9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
double	Double-precision floating point	4.9e-324 to 1.8e+308
float	Single-precision floating point	1.4e-45 to 3.4e+38
boolean	True/ False value	-
char	Character	-

Data Types (Contd)

Primitive Data Types Example:

```
byte value = 20;
short s1 = 1000;
long var = 2351678;
int a = 10;
float b = 1.56f;              // f specifies float value
double d1 = 12.6543d;         // d specifies double value
char ch = ‘a’;                // Enclosed by single quotation
boolean answer = true;
```

Data Types (Contd)

Non-primitive Data Types Example:

- **String**: A sequence of characters surrounded by **double quotes.**
  - **Example**:
    String my_string = “Hello World”;

- **Arrays**: Collection of a fixed number of values of similar data type.
  - **Example:**
    int [ ] arr1 = {24, 9, -11, 5};

THANK YOU!